MOBILE AD HOC NETWORKS (MANETs)

Outline

- Mobile Ad hoc networks.
- Applications of ad hoc networks.
- Challenges in Ad hoc network.
- Media Access Control
- Routing algorithms for MANETs.
- Conclusion.

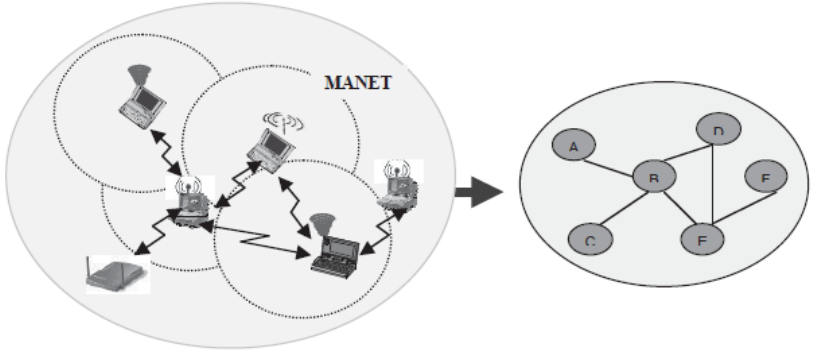
Introduction

- An **ad hoc network** is a **network** that is composed of individual devices communicating with each other directly.
- The term implies spontaneous or unprepared construction because these **networks** often bypass the gatekeeping hardware or central access point such as a router.

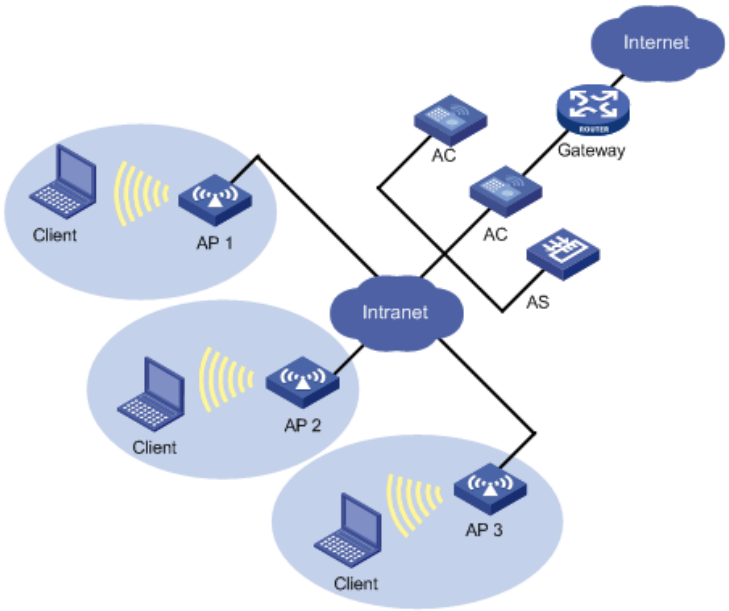
Mobile Ad hoc networks (MANETs)

- Mobile ad hoc networks are formed dynamically by an autonomous system of mobile nodes that are connected via wireless links.
- No existing fixed infrastructure or centralized administration – No base station.
- Mobile nodes are free to move randomly.
  - Network topology changes frequently.
- May Operate as standalone fashion or also can be connected to the larger internet.
- Each node work as router.

MANETs



MANETs



WLAN

Applications

- Tactical networks
  - Military communication, automated battlefields
- Emergency Services
  - Search and rescue operations
  - Disaster recovery – Earthquakes, hurricanes .
- Educational
  - Virtual classrooms or conference rooms.
  - Set up ad hoc communication during conferences, meeting, or lectures
- Home and Entertainment
  - Home/office wireless networking.
  - Personal Area network
  - Multiuser games
  - Outdoor internet access.

Challenges

- Infrastructure less
  - Brings new network designing challenges.
- Dynamically changing topologies
  - Cause route changes, frequent network partitions and packet loss.
- Physical layer limitations
  - Limited Wireless range.
  - Packet loss during transmission.
  - Broadcast nature of the communication.
- Limitations of Mobile Nodes
  - Short battery life
  - Limited capacities.
- Network security.

Effects on the protocol stack

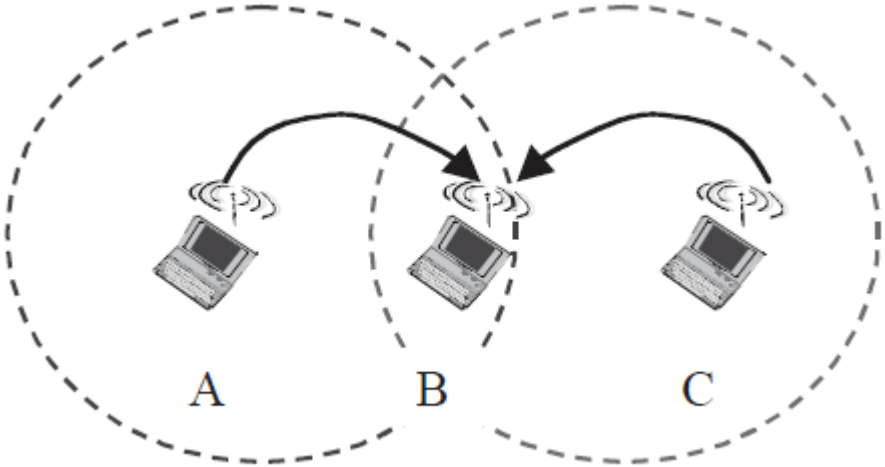
- Application Layer
  - New applications, Authentication, Encryption.
- Transport Layer
  - Congestion Control, Flow control.
- Network
  - Host addressing, Routing, Multicasting.
- Data Link Layer
  - Media Access
- Physical
  - Spectrum usage/allocation

Media Access Control

- Since MANETs, use broadcasting and shared transmission media, introduces a probability of packet collisions and media contention.
- Since collision detection is not possible with half-duplex radio. This brings new challenges to conventional CSMA/CD-based and MAC based protocols.
- Among the top issues are the hidden-terminal and exposed-terminal problems.

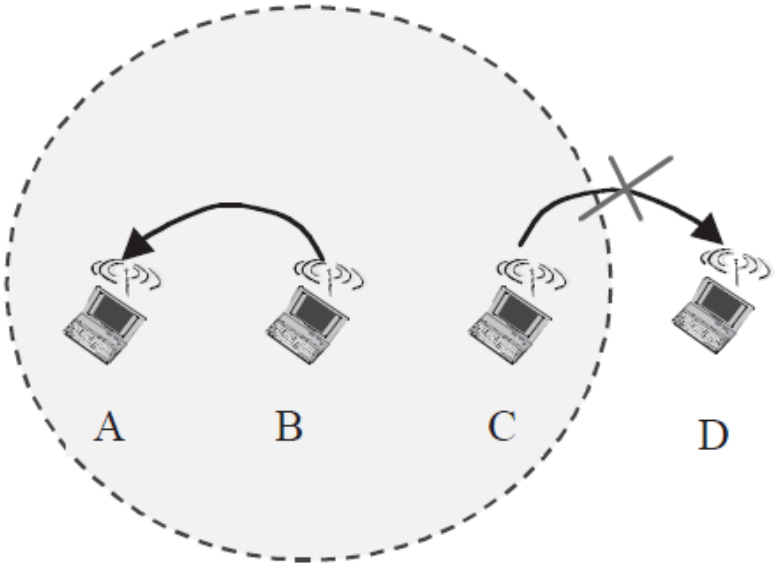
Hidden-terminal problem

- When two terminals can not detect each other's transmission due to being outside of each others range. The collision can occur.



Exposed-terminal problems.

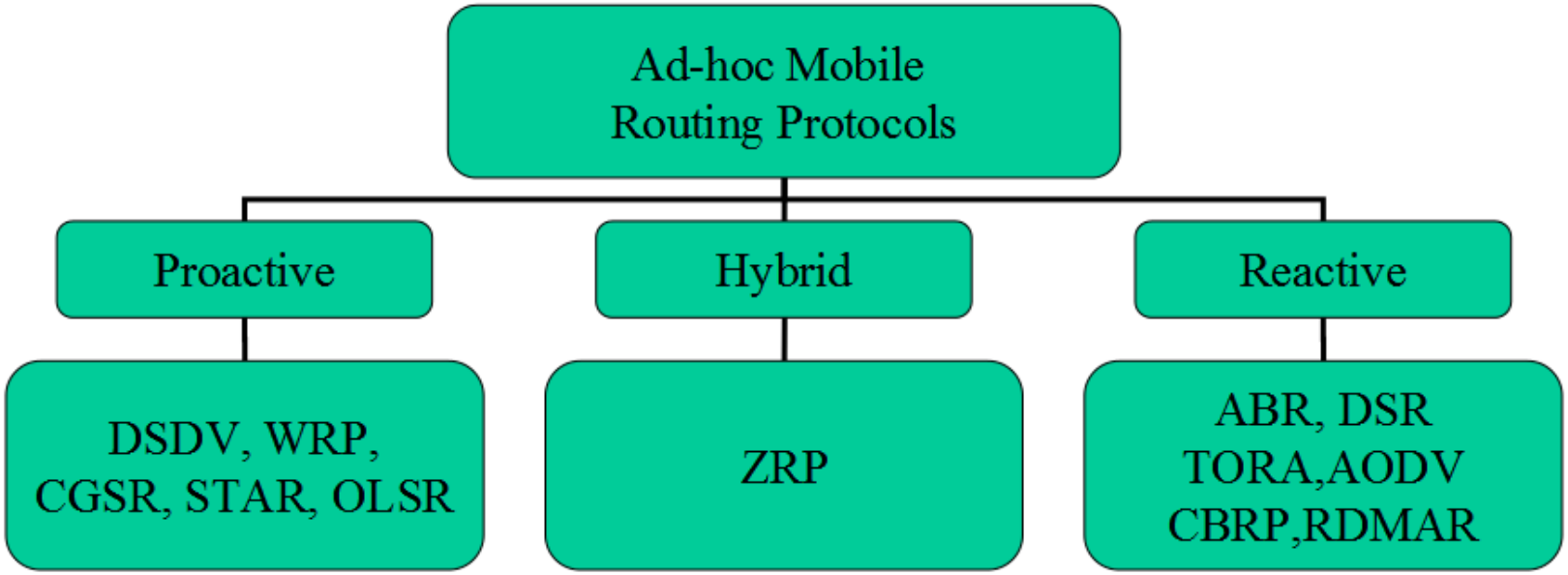
- Occur when a permissible transmission from a node to another node has to be delayed due to the irrelevant transmission between two other nodes.



Solution

- A new protocol **MACA** (multiple access with collision avoidance protocol) is used to avoid the Hidden-terminal and Exposed-terminal problems.
- Use signalling packets to avoid collision.
  - RTS (Request to send)
    - Sender request the right to send from a receiver with a short RTS packet before it sends a data packet.
  - CTS (Clear to send)
    - Receiver grants the right to send as soon as it is ready to receive

Routing In MANETs



Routing in Ad Hoc Networks

- Mobile nodes operate as routers.
- Proactive Protocol – Table Based
  - Maintain routes between every host pair all the time
  - Shortest-path protocols
  - Frequently update routing table; High routing overhead.
- Reactive Protocols – On-Demand
  - On-demand
  - Source initiates route discovery.
- Hybrid protocols
  - Combination of proactive and reactive.

Destination-Sequenced Distance-Vector (DSDV)

- Adapted from Routing Information Protocol(RIP).
  - Adds new attribute- **Sequence Number**
- Each node maintains a routing table which stores
  - Next hop
  - Cost matric for each destination
  - A sequence number that is create by destination itself.
- Each node advertises a monotonically increasing even sequence number.
  - Used to update path to destination node.
  - Odd sequence number used to represent broken path.
- Routing table updates are transmitted periodically.
- Routing information is transmitted by broadcast.

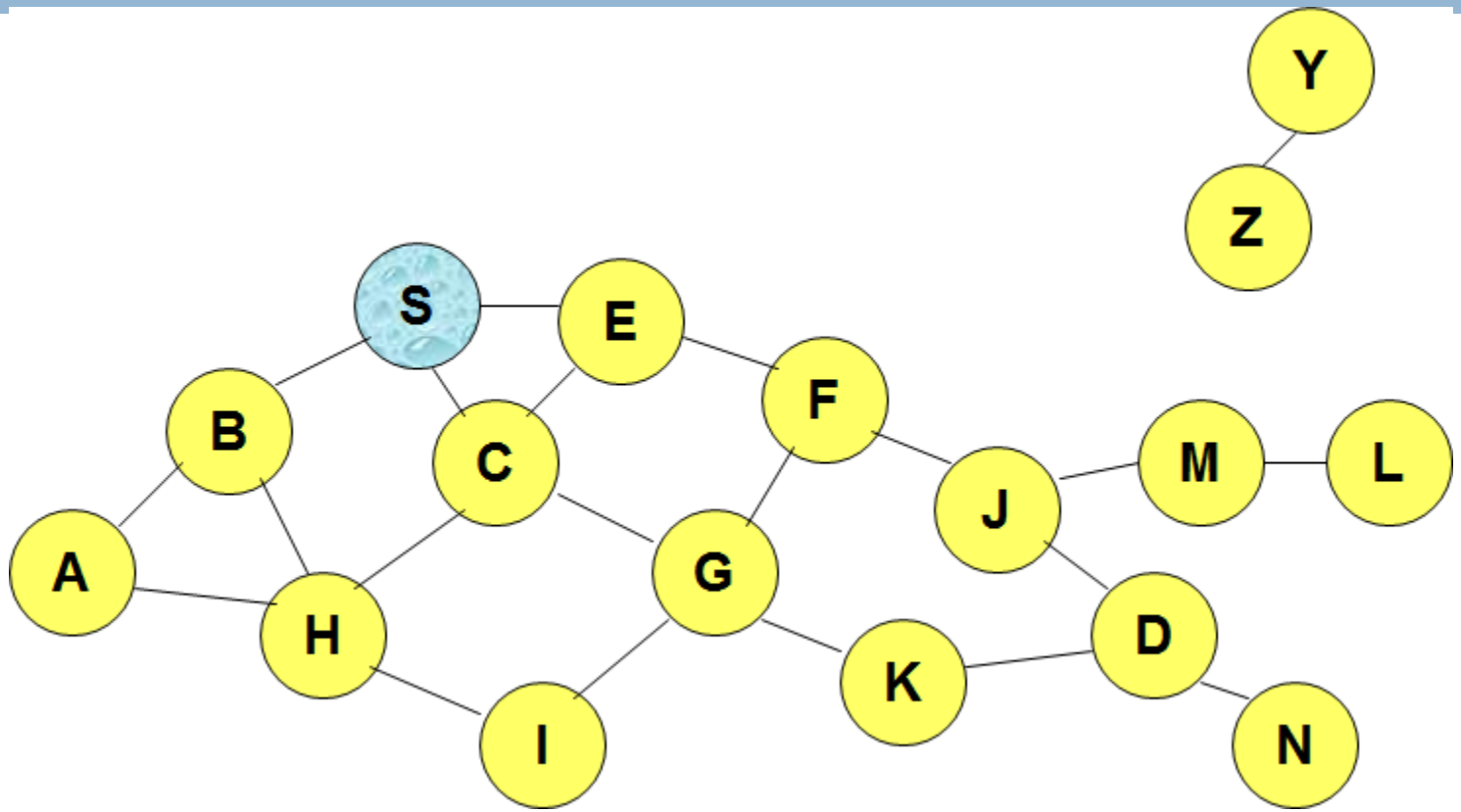
Dynamic Source Routing (DSR)

- DSR is a type of **Reactive Routing Protocols.**
- **On demand** Protocol
- Source node initiate **route discovery**.
- Source node floods **Route Request (RREQ)** in the network.
- Use broadcast method to send RREQ.
- Route request use a sequence number and path it traversed.
  - Sequence number is used to identified the request; to avoid looping.

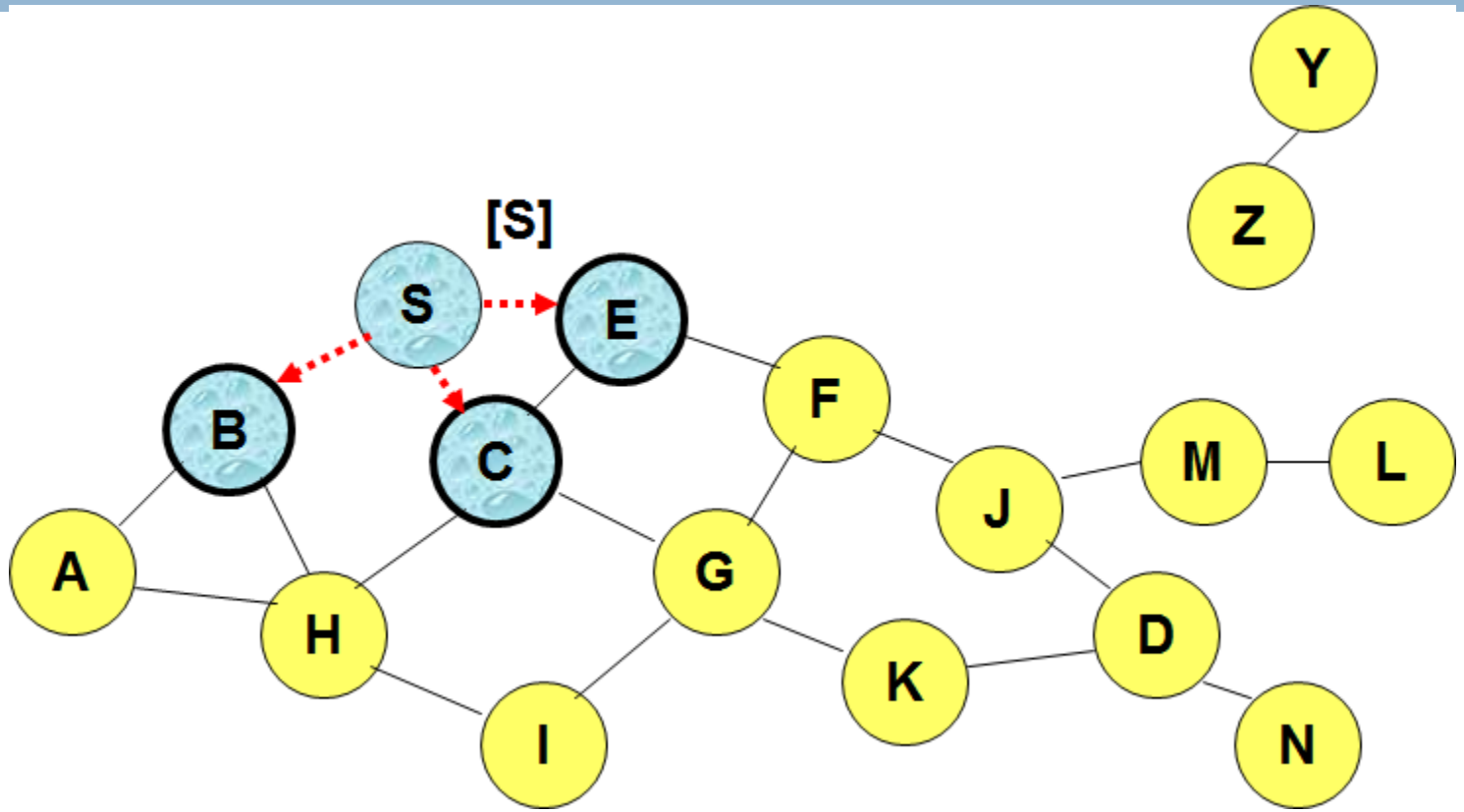
Cont'd

- The indented receiver sends a **Route Reply (RREP).**
- The receiver uses the reverse order of path in the Route Request to send the RREP.
- Source use received path to send data.
- Route may saved in source`s cache for future use.

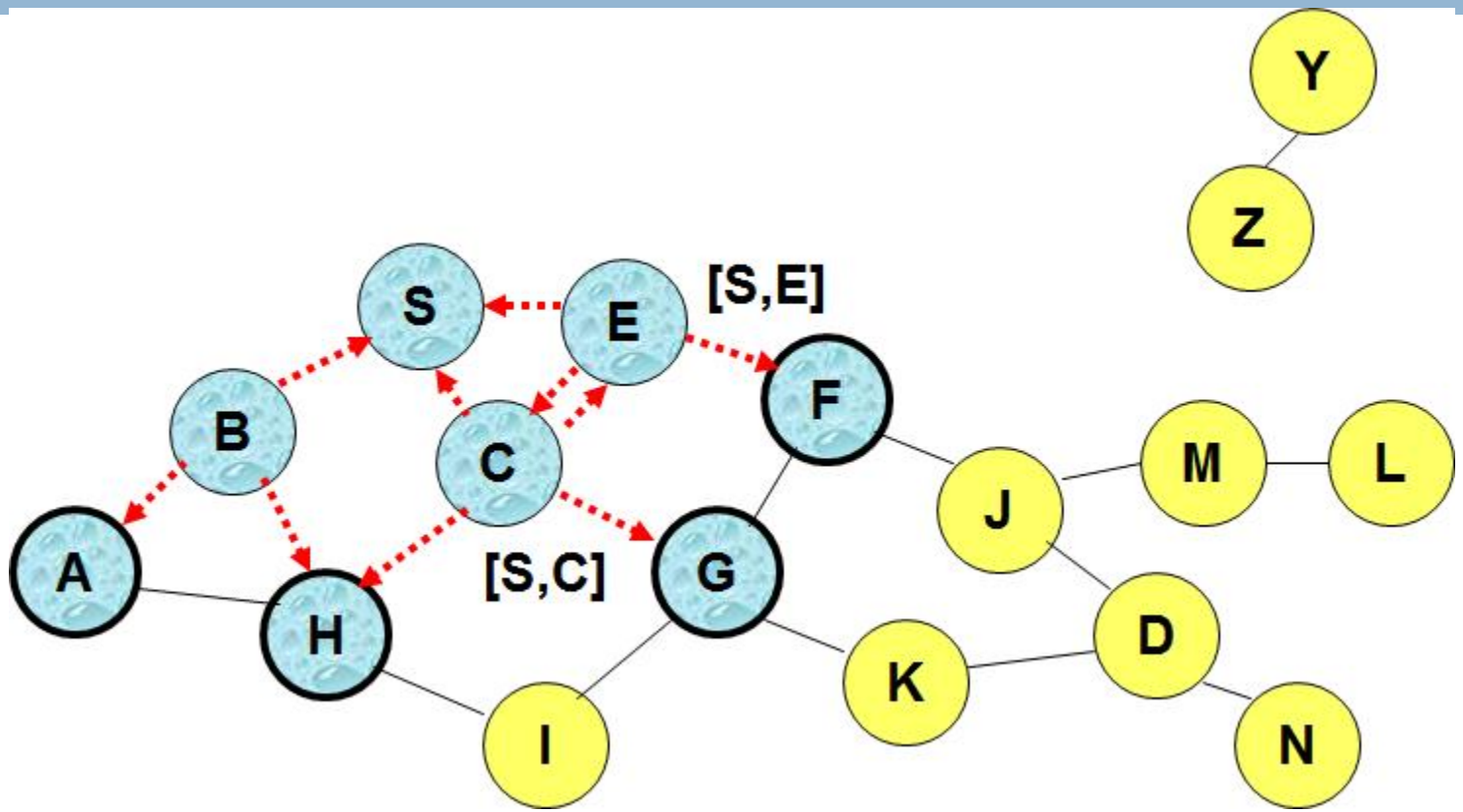
DSR



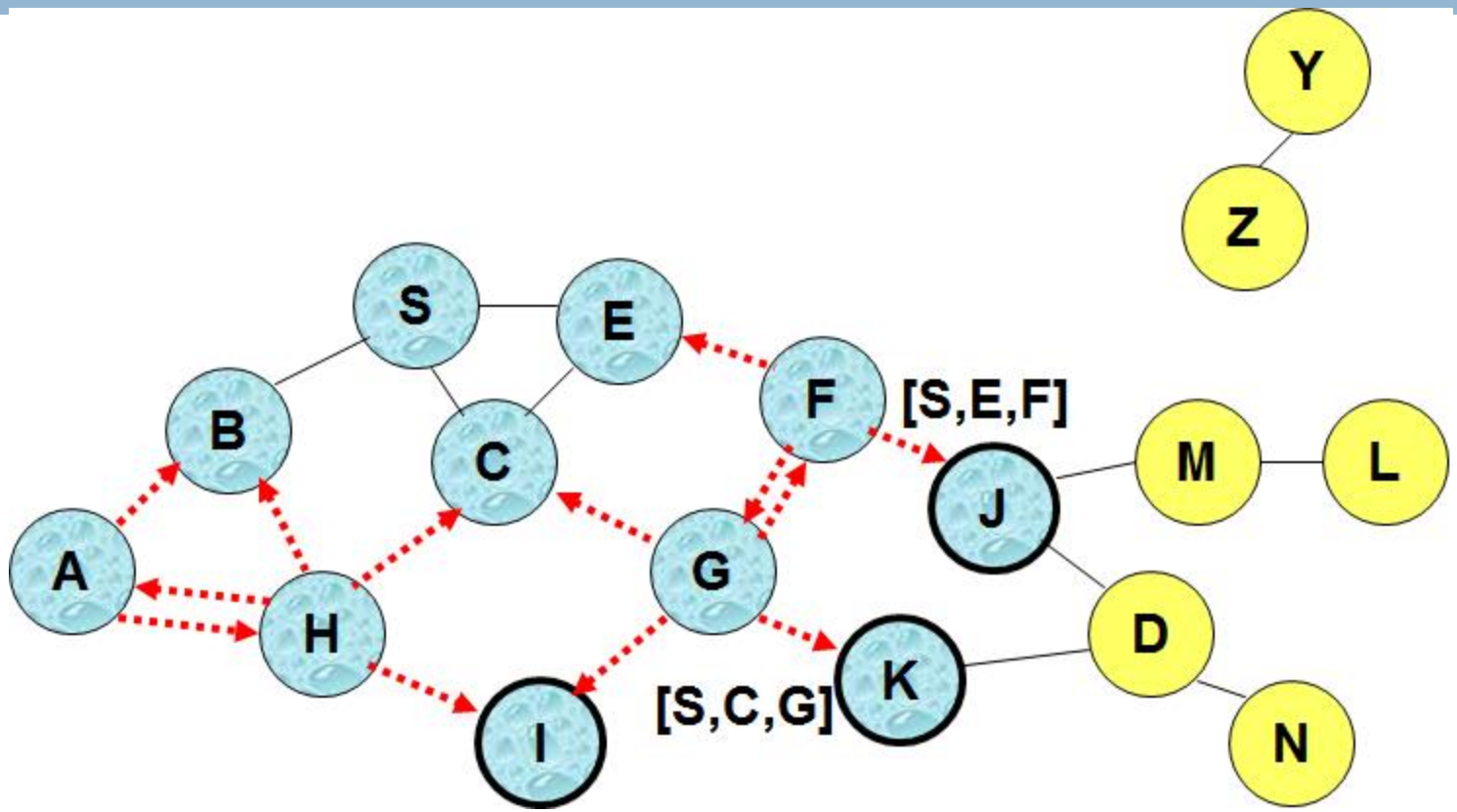
DSR



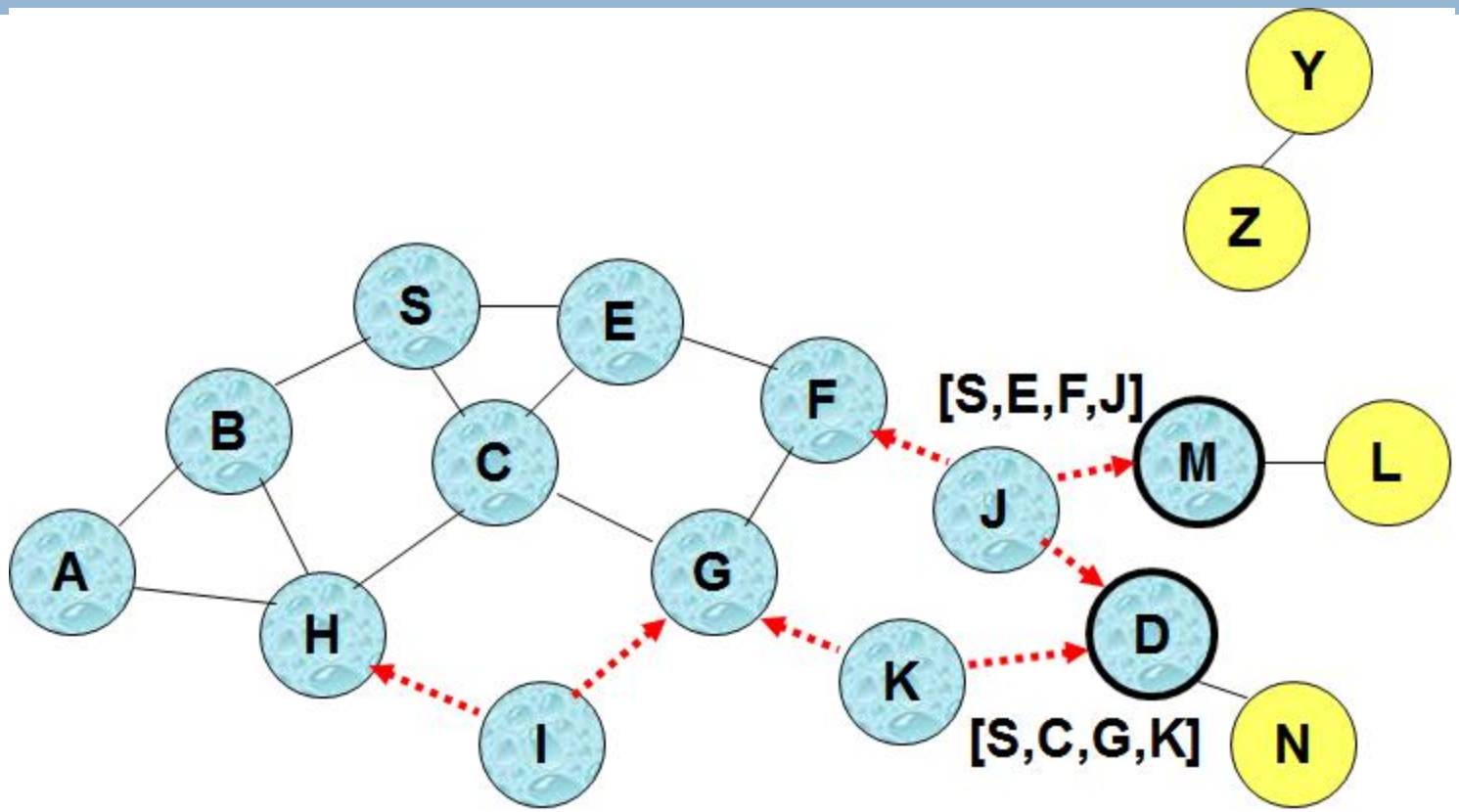
DSR



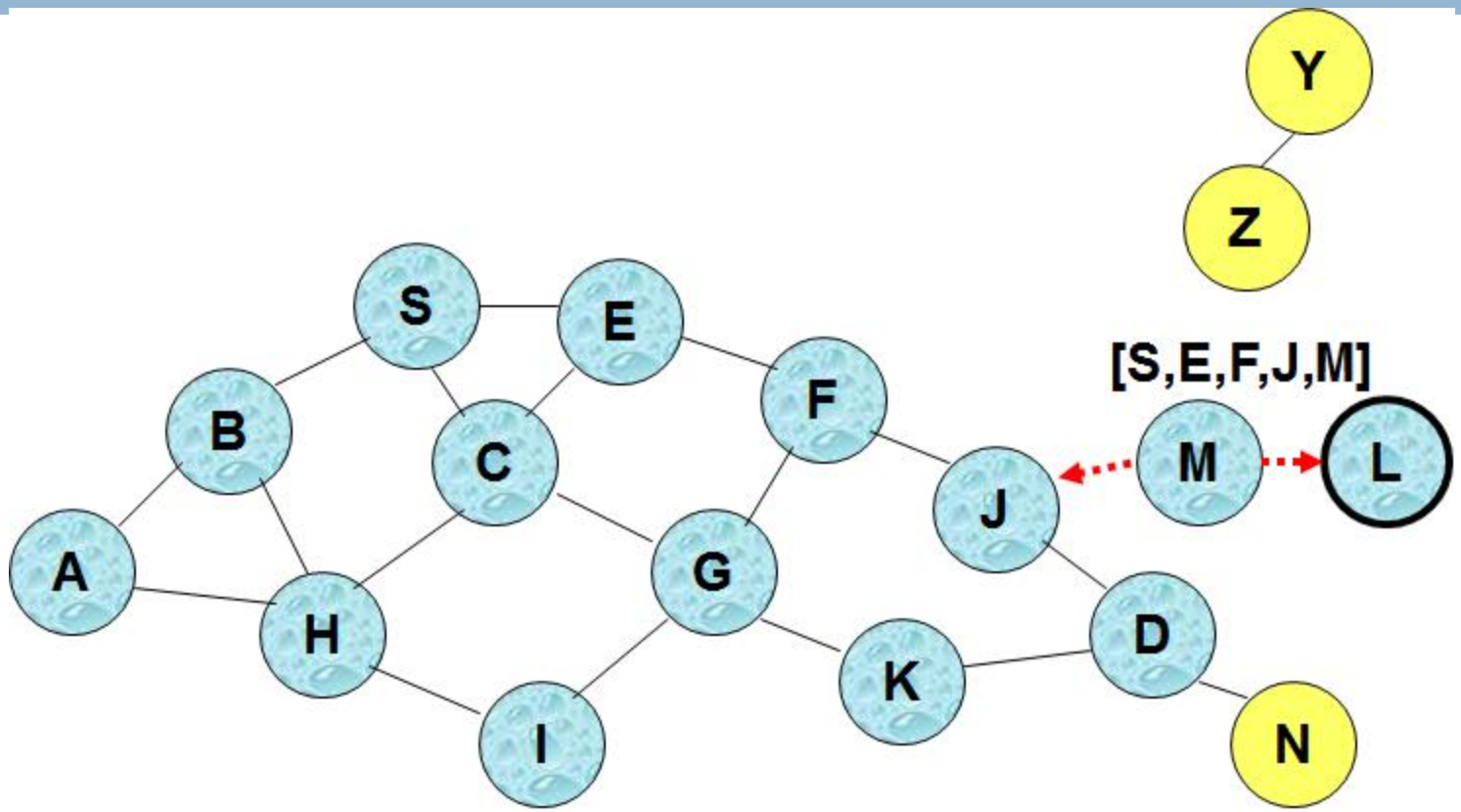
DSR



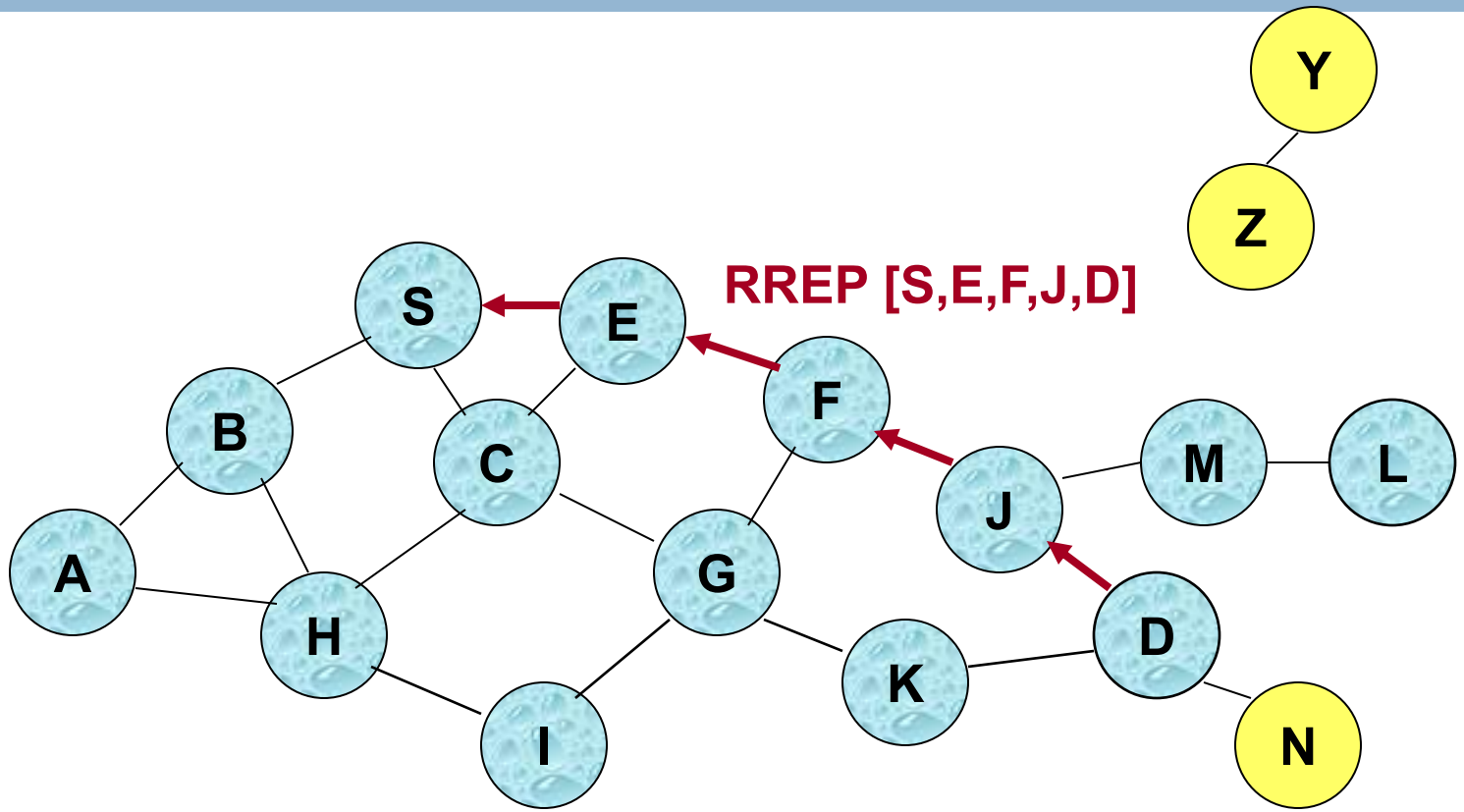
DSR



DSR



DSR (Reply)

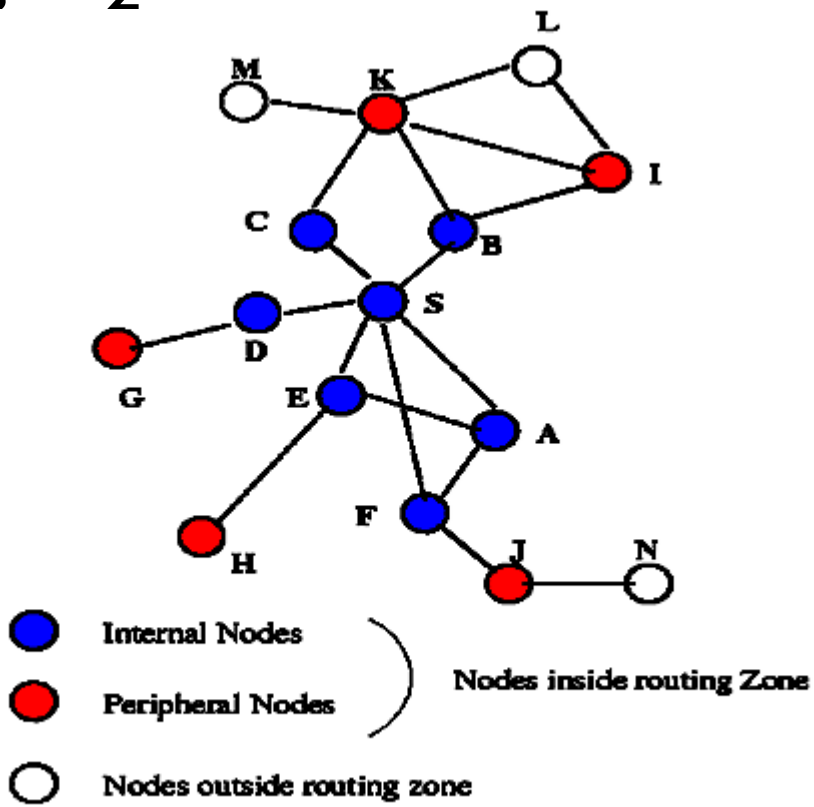


Zone Routing Protocol

- A Hybrid Protocol
- ZRP combines both proactive and reactive approach.
- Divide network into non- overlapping routing zones and use independent protocol on different zones.
  - **Intra-zone:** Proactive protocols are used.
  - **Inter-zone:** Reactive protocols are used.
- Routing zone diameter selected that optimized the routing.

Zone Routing Protocol

- Zone radius = 2



The End

References

- Basagni, Stefano, Marco Conti, Silvia Giordano, Ivan Stojmenović, et al. *Mobile Ad Hoc Networking*. Wiley-IEEE Press, 2004. Print.
- Iyer, Sridhar. *Mobile Ad Hoc Networks*. Bombay: 2000. Web. <www.it.iitb.ac.in/~sri/talks/manet.ppt>.